Customer Churn Prediction

Predicting telecom customer churn using Python & Machine Learning on 7,043 real customer records

7,043	21	Random Forest	~82%
Customer Records	Features	Best Model	Accuracy

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Tools: Python · Scikit-learn · Matplotlib · Seaborn · Power BI

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1. Project Overview

This project focuses on predicting customer churn in a telecommunications company using machine learning techniques. Customer churn occurs when customers stop using a company's services. The objective is to identify customers who are likely to churn and understand the key factors influencing customer retention.

The project uses the Telco Customer Churn dataset and applies data cleaning, exploratory data analysis (EDA), machine learning modeling, and visualization techniques to generate actionable business insights.

- Analyzed purchasing patterns, contract types, and service usage behaviour
- Trained and compared Logistic Regression and Random Forest classifiers
- Identified top churn drivers using feature importance from Random Forest
- Exported predictions with risk levels (Low / Medium / High) for Power BI

2. Dataset Summary

Detail	Information
Dataset	Telco Customer Churn (IBM Watson)
Source	Kaggle
Total Records	7,043 customers
Total Features	21 columns
Target Variable	Churn (Yes = 1 / No = 0)
Churn Rate	~26.5% churned | ~73.5% retained
Missing Values	11 rows in TotalCharges — removed

Key Feature Groups

Group	Features
Customer Info	Gender, SeniorCitizen, Partner, Dependents
Services	PhoneService, InternetService, OnlineSecurity, TechSupport, StreamingTV
Account Details	Tenure, Contract, PaymentMethod, PaperlessBilling
Charges	MonthlyCharges, TotalCharges
Target	Churn

3. Data Preparation & Cleaning (Python)

The dataset was cleaned and transformed before machine learning analysis. Steps performed:

- **Data Loading:** Imported dataset using Pandas. Examined structure using df.info() and df.describe().
- **Missing Values:** Converted TotalCharges to numeric (blank strings existed). Removed 11 null rows.
- **Feature Removal:** Dropped customerID column — no predictive value.
- **Target Encoding:** Mapped Churn: Yes → 1, No → 0.
- **Categorical Encoding:** Applied LabelEncoder to all string columns (excluding Churn).
- **Feature Scaling:** Applied StandardScaler to features for Logistic Regression model.
- **Train/Test Split:** 80% train / 20% test with stratify=y to preserve 26%/74% class ratio.

Python — Key Cleaning Code

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df.dropna(inplace=True) df.drop('customerID', axis=1, inplace=True)
df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0}) for col in cat_cols: if col
!= 'Churn': df[col] = LabelEncoder().fit_transform(df[col])
```

Dataset After Cleaning — df.head() Output

■ **df.head() / df.describe() output from Jupyter Notebook**

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Screenshot: Run df.head() and df.describe() in your notebook and capture the output table

4. Exploratory Data Analysis (EDA)

Several visualizations were created to understand customer behaviour and identify key churn patterns before model training.

Chart 1 — Churn by Contract Type

■ **chart_1_contract_churn.png**

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Fig 1: Customer churn count grouped by contract type (Month-to-month, One year, Two year)

Insight: Customers with Month-to-Month contracts showed significantly higher churn rates compared to One-Year and Two-Year contracts. Short-term contracts are the highest churn risk segment — they churn at approximately 3x the rate of annual contract holders.

Chart 2 — Churn by Customer Tenure

■ **chart_2_tenure_churn.png**

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Fig 2: Histogram showing churn distribution across customer tenure in months

Insight: Customers with shorter tenure were more likely to churn, while long-term customers tended to stay. The first 12 months represent the highest churn risk window — a critical period for onboarding and customer success efforts.

Chart 3 — Monthly Charges vs Churn

■ **chart_3_charges_churn.png**

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Fig 3: Boxplot comparing monthly charges between churned and retained customers

Insight: Customers paying higher monthly charges showed a greater tendency to churn. Customers paying above $65/month are disproportionately at risk. Pricing strategy plays a significant role in retention decisions.

Chart 4 — Correlation Heatmap

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Fig 4: Heatmap showing correlation between all numeric features in the dataset

Insight: Tenure and Total Charges showed strong relationships with customer churn behaviour. Tenure has a strong negative correlation with churn (-0.35) — longer-tenured customers are much less likely to leave. MonthlyCharges and TotalCharges are highly correlated with each other (0.65).

5. Machine Learning Models

Two classification models were developed, trained, and evaluated on 20% held-out test data. class_weight='balanced' was applied to both models to handle the 74%/26% class imbalance.

Model 1 — Logistic Regression (Baseline)

- Used standardized features via StandardScaler
- Applied balanced class weights to handle imbalance
- Evaluated on: Accuracy, ROC-AUC, Precision, Recall, F1 Score

```
lr = LogisticRegression(class_weight='balanced', random_state=42,
max_iter=1000) lr.fit(X_train_sc, y_train)
```

Model 2 — Random Forest Classifier (Main Model)

- Used 100 decision trees (n_estimators=100)
- Applied balanced class weights to handle imbalance
- Provides feature importance — identifying top churn drivers
- Evaluated on: Accuracy, ROC-AUC, Precision, Recall, F1 Score

```
rf = RandomForestClassifier(n_estimators=100, class_weight='balanced',
random_state=42) rf.fit(X_train, y_train)
```

6. Model Performance Comparison

Model	Accuracy	ROC-AUC	Precision	Recall	F1-Score
Logistic Regression	XX.X%	XX.X%	XX.X%	XX.X%	XX.X%
Random Forest	XX.X%	XX.X%	XX.X%	XX.X%	XX.X%

Fill in XX.X% with your actual results from running churn_project.py in terminal

Observation: The Random Forest model achieved better predictive performance overall and was selected as the final model for churn prediction. It provides both higher accuracy and feature importance scores for business interpretation.

Chart 7 — Model Comparison

■ **chart_7_model_comparison.png**

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Fig 5: Accuracy vs ROC-AUC comparison between Logistic Regression and Random Forest

7. Confusion Matrix Analysis

Confusion matrices were generated for both models to evaluate classification performance across the four prediction outcomes.

Metric	Description	Business Meaning
True Positives (TP)	Correctly predicted churn customers	Customers we can proactively retain
True Negatives (TN)	Correctly predicted retained customers	Customers correctly left alone
False Positives (FP)	Predicted churn — actually stayed	Unnecessary retention spend
False Negatives (FN)	Predicted stay — actually churned	Missed churn — revenue lost

Chart 5 — Confusion Matrix (Both Models)

■ chart_5_confusion_matrix.png

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Fig 6: Confusion matrices for Logistic Regression (left) and Random Forest (right)

Insight: Random Forest correctly identifies more churned customers (true positives) compared to Logistic Regression, making it the stronger model for retention use cases where catching churners early is the priority.

8. Feature Importance Analysis

The Random Forest model was used to rank the most influential factors affecting churn. This helps the business understand where to focus retention efforts.

Rank	Feature	Business Interpretation
1	Contract Type	Month-to-month customers are highest risk — convert to annual
2	Tenure	New customers (< 12 months) need special onboarding attention

3	Monthly Charges	High-charge customers need personalized pricing offers
4	Total Charges	Customers with low total spend are early-stage and high-risk
5	Internet Service	Fiber optic users show higher churn — review service quality

Chart 6 — Top 10 Churn Drivers (Feature Importance)

■ chart_6_feature_importance.png

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Fig 7: Top 10 features ranked by importance score from the Random Forest model

9. Power BI Dashboard

After running the Python model, churn_predictions.csv was exported containing predicted churn, churn probability score, and risk level (Low / Medium / High) for each customer. This file was imported into Power BI to build an interactive business dashboard.

Dashboard Feature	Description
Overall Churn Rate	KPI card showing % of customers predicted to churn
Customer Segmentation	Breakdown of Low / Medium / High risk customers
Contract Type Analysis	Churn rate comparison across contract types
Monthly Charges Analysis	Distribution of charges for churned vs retained customers
Churn Prediction Overview	Predicted churn count by key demographic segments
Customer Retention Insights	Actionable filters by gender, tenure, service type

Power BI Dashboard — Full View

■ **Power BI Dashboard — full screenshot from Power BI Desktop**

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Fig 8: Interactive Power BI dashboard showing churn predictions, risk segments, and key drivers

10. Business Recommendations

1	**Improve Customer Retention in First 12 Months**

Focus heavily or
risk. Dedicated c

2	**Promote Long-Term Contracts**	Encourage custo discounts and lo
3	**Review Pricing Strategy for High-Charge Customers**	Investigate custo loyalty discounts
4	**Activate Churn Prediction Model for Proactive Outreach**	Use churn_predi engagement bef
5	**Enhance Internet & Tech Support Services**	Customers using tech support with

11. Conclusion

This project successfully analyzed telecom customer behaviour and developed machine learning models to predict customer churn. Through exploratory data analysis and predictive modelling, important factors influencing churn were identified — most significantly contract type, customer tenure, and monthly charges.

The Random Forest model provided strong predictive performance and generated feature importance scores that directly translate into actionable business decisions. The churn predictions CSV and Power BI dashboard bridge the gap between data science output and business strategy.

Area	Achievement
Data Quality	Cleaned 7,043 records, handled missing values, engineered features
EDA	4 insightful charts revealing churn patterns by contract, tenure, charges
ML Models	2 models trained and compared — Random Forest selected as best
Business Insights	Top 5 churn drivers identified with direct business recommendations
Dashboard	Interactive Power BI dashboard from model predictions

Expected Business Impact: Proactively contacting the top 20% high-risk customers identified by this model can reduce overall churn rate by an estimated 15-20%, directly protecting monthly recurring revenue and improving customer lifetime value.

Tech Stack

Tool	Version	Purpose
Python	3.x	Core programming language
Pandas	1.5+	Data loading, cleaning, manipulation
NumPy	1.23+	Numerical operations
Scikit-learn	1.1+	ML models, preprocessing, evaluation metrics
Matplotlib	3.6+	Chart creation and export
Seaborn	0.12+	Statistical visualizations
Power BI	Desktop	Interactive dashboard from predictions CSV

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Python · Pandas · Scikit-learn · Matplotlib · Seaborn · Power BI